

Polun Wu

polunwu@gmail.com • [Linkedin](#) • [Github](#) • [Website](#)

Summary

Software Engineer with **5+ years of experience** building **product-grade web application**. Passionate about **user experience** and focusing on **building scalable frontend systems** and **improving engineering efficiency**. Specialized in **frontend architecture**, **performance optimization**, **testing strategies** and **UI consistency at scale**.

Experience

Software Engineer - GILOO Group Limited

Remote, Taipei • 09/2021 - 04/2026

- Introduced a **modular architecture** for the **video player UI**, facilitated the retirement of legacy UIs with independently **maintainable modules** that ensured consistent patterns, enabled localized changes, reduced browser compatibility issues and significantly **improved playback feature productivity**.
- Established a testing mechanism by **introducing UI end-to-end testing** for **critical video content entitlement and playback state flows**, defining a structured test plan and scenarios. Successfully **reduced P0 production bugs by 50%** and enhanced **future release confidence**.
- Optimized **performance on critical pages** and improved **SEO**. Enhanced loading performance by **splitting JavaScript bundles**, establishing well-structured **page hierarchy**, eliminating **layout shifts**, **reducing load times by 56%** with **LCP under 2s** and **CLS of 0.00**, resulting in **increased organic traffic approximately 1.3 times** and **improved product visibility**.
- Implemented a **server-side LRU caching strategy** to optimize SSR performance, **reducing CPU load** during server-side rendering and **improving overall request efficiency**.
- **Integrated DRM license workflows** for protected video playback, ensuring reliable playback compatibility across modern browsers.
- Built **AI-powered translation features** for **video subtitles** and **metadata** in a **content creator dashboard**. Collaborated on **project planning**, outlined **execution details**, and delegated tasks to one engineer. Thereby reducing **localization effort** and friction in content publishing workflows, while contributing to **increased creator adoption** and **higher content output**.
- Led modern **frontend architecture initiatives** by **structuring, migrating, documenting a well-layered architecture** across UI, feature logic, state management, API layer, and form schema validation, standardizing development practices and **improving scalability, maintainability, and developer experience** across the product.
- Leadership & collaboration: mentored a new engineer and improved team onboarding processes by **documenting standardized onboarding materials** and **code review guidelines**, enhancing collaboration efficiency, code quality and engineering consistency across the team.

Tech Stack: TypeScript, JavaScript, React.js, Next.js, TanStack Query, RTK Query, Zod, Zustand, Tailwind CSS, React Hook Form, MUI, shadcn/ui, Vue.js, Nuxt.js, Vuex, Vuetify, Node.js, Docker, Gitlab CI, Cypress, Jest, ESLint, Prettier, Husky

Frontend Engineer - Polish Design

Remote, Taipei • 02/2020 - 09/2021

- Developed **browser-based 3D visualizations** and mobile **augmented reality (AR) interactive experiences**.
- Designed and developed **interactive web animations** using **GSAP** and **CSS animations**, creating **immersive user experiences** across multiple client products.
- Conducted research and built **interactive prototypes** to help UX validate design feasibility, **reducing development risks** before implementation.
- Contributed to the redesign of an **enterprise network management platform** by introducing **modern frontend frameworks** and a **standardized UI component library**, improving **scalability** and **development consistency**.
- Built a **hybrid Vue.js/Rails micro-frontend** for an **e-commerce platform**, architecting a **centralized cart state** synchronized across product discovery, cart, checkout, discount, and shipping workflows.

Tech Stack: JavaScript, CSS Animations, SCSS, GSAP, Three.js, Vue.js, Nuxt.js, Vuex, REST APIs, Ruby on Rails

Projects

Golden Book RAG	2026
- Implement a retrieval-augmented generation (RAG) based question answering system for PDF documents, enabling users to query books with responses grounded in block-level citations. - Built from-scratch: parsing, chunking, embeddings, semantic search, evaluation, and grounded LLM generation with citations and anti-hallucination safeguards. Tech Stack: Python, FastAPI, PostgreSQL, pgvector, Gemini, Next.js Demo • Github	
google-style-code-review-skills	2026
- A Claude Code skill for code reviews following Google's Engineering Practices guidelines. - Produced standardized comments severity labels, e.g. blocking, nit, optional, informational. Aimed at introducing a consistent evaluation approach to improve generated code quality. Github	

Skills

Front-end: TypeScript, JavaScript, React.js, Next.js, TanStack Query, Zustand, Tailwind CSS, shadcn/ui, Styled Components, React Hook Form, Vue.js, Vue3, Jest, Cypress Back-end: Python, Golang, PostgreSQL, pgvector Tools: Git, Chrome DevTools, Lighthouse, Figma, Storybook, Google Search Console, Vite, ESLint, Prettier DevOps: GitLab CI/CD, GitHub Actions, Docker/Kubernetes	
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AI Methodology

Hands-on experience with human-AI collaboration by adopting harness engineering, rule-centric, documentation-driven AI-assisted development. Leverage agentic workflow across the software lifecycle, from research and prototyping to implementation, refactoring, growth and maintenance. Grounding AI agent with domain knowledge to amplify productivity and code quality.

Education

M.S. in Mechanical Engineering - National Central University	2016 - 2019
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Research & Awards

Customized PEEK Implants With Microporous and Surface Modification Using 3D Printing 2nd Place Best Paper — 2019, ASME 13th International Conference on Micro and Nano Systems	2019
Process Optimization of PEEK 3D Printing Using Taguchi Methods 3rd Place — 2018, Chinese Society of Mechanical Engineers Conference 35th	2018